Controllability of Multi-Term Fractional-Order Impulsive Dynamical Systems with phi-Caputo Derivative

Md Samshad Hussain Ansari and Muslim Malik

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Abstract

This article examines the exact controllability of multi-term fractional differential equations with impulses and the generalized phi-Caputo fractional derivative. By utilizing generalized Mittag-Leffler functions and Krasnoselskii fixed-point methodology, rigorous controllability criteria are established for impulsive dynamical systems.

1. Introduction

Multi-term fractional differential equations model multi-rate relaxation processes in viscoelastic fluids, dielectric relaxation, and anomalous subdiffusion with multiple power-law regimes. Impulsive effects represent abrupt state jumps occurring at discrete time instants t_k, widely observed in pharmacokinetics and power electronics.

The phi-Caputo fractional derivative, introduced by Almeida, unifies classical Caputo, Caputo-Katugampola, and Caputo-Hadamard derivatives through an increasing monotonic function phi(t).

2. Preliminaries on phi-Caputo Calculus and Impulsive Systems

Let phi in C^1([0, T], R) be a strictly increasing function such that phi'(t) != 0 for all t in [0, T].

Definition 2. *1 (phi-Caputo Derivative). For alpha > 0, the phi-Caputo fractional derivative of f is defined by ^C D^{alpha; phi}_t f(t) = (1 / Gamma(n - alpha)) int_0^t (phi(t) - phi(s))^{n - alpha - 1} f^{[n]}_phi(s) phi'(s) ds, where f^{[n]}_phi(s) = ((1 / phi'(s)) d/ds)^n f(s).*

Consider the multi-term impulsive fractional dynamical control system:

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^C D^{alpha; phi}_t x(t) + sum_{i=1}^m c_i ^C D^{alpha_i; phi}_t x(t) = A x(t) + B u(t) +
                  f(t, x(t)), t != t_k, Delta x(t_k) = I_k(x(t_k^-))
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Lemma 2. *2 (Impulsive Cauchy Problem Mild Solution). A piecewise continuous function x in PC([0, T], X) is a mild solution if and only if it satisfies the integral representation involving multi-term characteristic Mittag-Leffler solution operators S_phi(t, s) and impulsive jump sums sum_{k: t_k < t} S_phi(t, t_k) I_k(x(t_k^-)).*

Assumption 3. *1 (Impulse Jump Compactness). The impulsive jump operators I_k: X -> X are continuous and satisfy ||I_k(x) - I_k(y)|| <= c_k ||x - y|| with sum_{k=1}^p c_k < 1.*

3. Main Controllability Theorems

Define the multi-term controllability Gramian operator W_phi = int_0^T S_phi(T, s) B B* S_phi*(T, s) phi'(s) ds. We assume W_phi is invertible.

Theorem 3. *2 (Exact Controllability of Multi-Term Impulsive Systems). Let Assumptions 3.1 and 3.2 hold, and let W_phi^{-1} exist as a bounded operator. If the compactness condition and contractive coefficient L_comp = ||S_phi|| [L_f T + sum c_k] (1 + ||B|| ||W_phi^{-1}|| ||S_phi||) < 1 is satisfied, then the multi-term impulsive dynamical system is exactly controllable on [0, T].*

Corollary 3. *3 (Single-Term Non-Impulsive Case). When c_i = 0 and I_k = 0 for all k, Theorem 3.2 reduces to standard exact controllability for phi-Caputo fractional systems.*

Remark 4. *1. Selecting phi(t) = t yields classical Caputo controllability; choosing phi(t) = ln(t) recovers Caputo-Hadamard fractional controllability with logarithmic memory decay.*